

Homework 2

CENG 113, SP 2026

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Academic Integrity: For all problems **show your work** to receive credit. By submitting your assignment you're confirming that your submission is a product of your own effort.

Box all answers!

Problem 1

Consider the following stoichiometric table:

Species	ν_{i1}	ν_{i2}	ν_{i3}
A	2	-3	-2
B	3	-2	-3
C	-3	0	3
D	3	3	0
E	-1	3	2

- (1.A) For each species write the net rate of reaction (r_i) in terms of ν_{ij} and r_j (e.g., $r_F = -r_1 + 2r_3$).
- (1.B) Assume reactions 1 and 2 are irreversible, and reaction 3 is reversible. For each reaction,
- Write the reaction (e.g., $A \rightleftharpoons 2B$)
 - Write the corresponding elementary rate law in terms of concentrations (e.g., $r_j = k_j C_A$).

Problem 2

Use the information provided to calculate the rate constant k at 65 °C. Clearly indicate the units of k .

- (2.A) $A = 10^5 \text{ s}^{-1}$, $E_a = 30 \text{ kJ} \cdot \text{mol}^{-1}$. (ANS: $k \approx 2.3 \text{ s}^{-1}$)
- (2.B) $A = 12\,345 \text{ L} \cdot \text{mol}^{-1} \cdot \text{h}^{-1}$, $E_a = 3 \text{ kcal} \cdot \text{mol}^{-1}$.
- (2.C) $k = 0.14 \text{ min}^{-1}$ at 30 °C with $E_a = 2.6 \times 10^4 \text{ J} \cdot \text{mol}^{-1}$. (ANS: $k \approx 0.4 \text{ min}^{-1}$)
- (2.D) $k = 6.2 \text{ m}^3 \cdot \text{kmol}^{-1} \cdot \text{s}^{-1}$ at 85 °C with $E_a = 14 \text{ kJ} \cdot \text{mol}^{-1}$.

Problem 3

Calculate the CSTR volume required to achieve 62% conversion of A according to the elementary, liquid-phase reaction



assuming the reactor operates at 75 °C. Additional information is provided below. (ANS: $V \approx 50 \text{ mL}$)

$$C_{A0} = 0.4 \text{ mol} \cdot \text{L}^{-1}$$

$$\nu_0 = 75 \text{ mL} \cdot \text{s}^{-1}$$

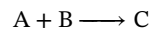
$$T_0 = 50 \text{ }^\circ\text{C}$$

$$k = 0.26 \text{ s}^{-1} \text{ at } 40 \text{ }^\circ\text{C}$$

$$E_a = 58 \text{ kJ} \cdot \text{mol}^{-1}$$

Problem 4

Ethylene oxide (A) is reacted with water (B) to form ethylene glycol (C, aka 1,2-ethanediol) according to the liquid-phase reaction



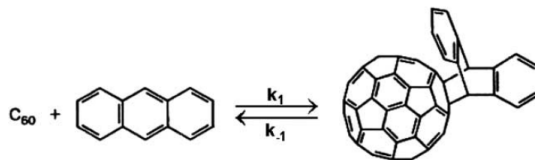
which follows a **non-elementary**, first-order rate expression $r = kC_A$ with $k = 0.03 \text{ min}^{-1}$ at 50 °C.

A solution containing 10 mol% ethylene oxide in water at 50 °C is provided to an isothermal, 0.75 m³ CSTR at a rate of 4 L·s⁻¹. If necessary, the densities of all components can be assumed equal to the density of water (1 kg·L⁻¹).

- (4.A) Calculate the conversion of ethylene oxide and outlet molar flow rate of ethylene glycol. (ANS: $n_C \approx 1.6 \text{ mol} \cdot \text{s}^{-1}$)
- (4.B) Suppose two utility fluids are available: liquid cooling water at 15 °C or superheated steam at 200 °C. Estimate the heat exchanger area A required maintain isothermal operation. The heat capacity and enthalpy of formation of ethylene oxide are 48 J·mol⁻¹·°C⁻¹ and -52.6 kJ·mol⁻¹. (ANS: $A \approx 7 \text{ m}^2$)
- (4.C) Suppose the reactor is operated adiabatically instead of isothermally. Determine the reactor temperature and outlet molar flow rate of ethylene glycol if the activation energy is 27 kJ·mol⁻¹. (ANS: $T \approx 75 \text{ }^\circ\text{C}$)

Problem 5

Those clever chemists¹ are always coming up with interesting synthetic processes, such as this one:



In this synthesis buckminster fullerene (C₆₀, or “buckyballs”) is functionalized with anthracene (C₁₄H₁₀) within an inert carrier liquid such as toluene. Exactly *why* one would be interested in doing such a thing is not addressed by the referenced paper but that’s the way it goes with fundamental research.

Sometimes the supply of anthracene contains small amounts of a contaminant molecule, tetracene (C₁₈H₁₂), which competes with anthracene for buckyballs according to the parallel Diels-Alder reactions



where D and U are the desired and undesired products. Sarova and Berberan-Santos found that the reactions above can be well-described as elementary reactions with the following kinetics:

Rate constant	A	E_a
$k_{1,f}$	$6.310 \text{ L} \cdot \mu\text{mol}^{-1} \cdot \text{s}^{-1}$	$59 \text{ kJ} \cdot \text{mol}^{-1}$
$k_{1,r}$	$7.943 \times 10^{13} \text{ s}^{-1}$	$140 \text{ kJ} \cdot \text{mol}^{-1}$
$k_{2,f}$	$25.12 \text{ L} \cdot \mu\text{mol}^{-1} \cdot \text{s}^{-1}$	$49 \text{ kJ} \cdot \text{mol}^{-1}$

A solution containing 10 000 μM anthracene,² 200 μM buckyballs, and 10 μM tetracene in toluene is provided to a CSTR at a rate of 1 mL·s⁻¹. The reactor operates isothermally at 62 °C with a residence time of 2000 min.

¹G.H. Sarova; M.N. Berberan-Santos. Kinetics of the Diels-Alder reaction between C60 and acenes. Chem. Phys. Lett. 397 (2004) 402-407.

²The unit “μM” is micromolar, or 10⁻⁶ mol·L⁻¹.

- (5.A) Calculate the volume of the reactor and compare your answer to an everyday object. Also calculate the volume of fluid which enters the reactor each second and compare your answer to an everyday object. (*hint: the calculation required for this problem is not lengthy*) (ANS: $V > 100 \text{ L}$)
- (5.B) Set up a stoichiometric table.
- (5.C) Determine the outlet concentrations (in μM) of all components except toluene (*note: to help solve converge you should express terms in each material balance with units of $\mu\text{mol}\cdot\text{s}^{-1}$*). (ANS: $C_D \approx 160 \mu\text{M}$)